

COMP9020

Foundations of Computer Science

2026 Term 3

- Textbook (R & W) - Ch. 1, Sec. 1.1-1.4
Ch. 2, Sec. 2.1
- Problem set 1
- Self-guided study: Exercises Ch. 1 (R & W), ...

COMP9020 26T3 Course Convenors

Name: Michael Thielscher
 Consults: **Fridays 2.30pm-3.30pm** (weeks 1–5,7–10)
 Room: **Room 310 in K17** (turn right from lift)
 Research: AI, Skilled Companions, Robotics, General Game Playing
 Pastimes: Fiction, Films, Food, Football

Email: cs9020@cse.unsw.edu.au for all course-related enquiries

Tutors: Ahmad Fauzaini
 Charran Kethees
 Damian Griffin
 Jiapeng Wang
 Joffrey Ji
 Varun Agarwal

Course Aims

The course aims to increase your level of mathematical maturity to assist with the fundamental problem of **finding, formulating, and proving** properties of programs.

The actual content is taken from a list of subjects that constitute the basis of the tool box of every serious practitioner of computing:

- numbers, sets, formal languages week 1
- logic, proofs, Boolean algebra week 2
- function and relation theory week 3–4
- graph theory week 5
- **mid-term test** (no lectures) week 6
- induction, recursion, order of growth, combinatorics week 7–8
- probability, expected values week 9–10

NB

“universitas” (Lat.) = sum of all things, a whole
 By acquiring knowledge and enhancing your problem-solving skills, you’re preparing yourself for the future

Course Material

All course information is placed on the WebCMS3 course website

webcms3.cse.unsw.edu.au/COMP9020/26T3/

Log in to access course materials.

Textbook:

- KA Ross and CR Wright: [Discrete Mathematics](#)

Supplementary textbook:

- E Lehman, FT Leighton, A Meyer: [Mathematics for Computer Science](#)

Lectures, Problem Sets, Quizzes

Lectures will:

- present theory
- demonstrate problem-solving methods

Lecture slides will be made available before lecture

Feel free to ask questions, but **No Idle Chatting**

The weekly **homework** and short assessments (**quizzes**) aim to:

- clarify any problems with lecture material
- work through exercises related to lecture topics

Homework (problem sets) made available before the lecture

Sample solutions posted the following week on Tuesdays.

Quizzes (from week 2): **in person, in your tutorial**

Do homework yourself! and **Don't fall behind!**



5

Assessment Summary

- 1 in-person tutorial quizzes (weeks 2, 3, 4, 5, 7, 8, 9, 10) — max. marks 14
- 2 in-person mid-term test (1 hour in week 6) — max. marks 26
- 3 in-person exam (2 hours in the exam period) — max. marks 60

NB

Your **overall score** for this course will be the **maximum** of

- tutorial + mid-term + exam
- tutorial + exam * $\frac{86}{60}$
- mid-term + exam * $\frac{74}{60}$
- exam * $\frac{100}{60}$

⇒ If you do better in the final exam, your weekly tutorial quizzes and/or mid-term test result will be ignored

⇒ The quizzes and mid-term test can only improve your final mark



6

NB

To pass the course, your overall score must be 50 or higher **and** your mark for the final exam must be 24 or higher ($\geq 40\%$).

The in-person quizzes

- are variations of clearly marked questions from the previous week's problem set
- are held at the beginning of your weekly tutorial
- are written on a sheet of paper given to you

You get your own individual questions for each quiz

- each quiz is worth 2 marks
- max. quiz mark = 14

We take your 7 best quiz results (of 8 quizzes)



7

Mid-term Test, Tutorials

NB

In-person test in week 6

(1 hour on Wednesday, 21 October, starting at 1pm)

You get your own individual test paper:

- some multiple-choice questions
- some descriptive/analytical questions with open answers

max. mid-term test marks = 26

Tutorials will:

- provide additional explanations and in-person exercises
- clarify any doubts about the homework

All tutorials start this week!



8

Numbers

Definition

Integers $\mathbb{Z} = \{\dots - 2, -1, 0, 1, 2, \dots\}$

Reals $\mathbb{R}$

$\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$ — **floor** of x , the greatest integer $\leq x$

$\lceil \cdot \rceil : \mathbb{R} \rightarrow \mathbb{Z}$ — **ceiling** of x , the least integer $\geq x$

Example

$$\lfloor \pi \rfloor = 3 = \lceil e \rceil \quad \pi, e \in \mathbb{R}; \lfloor \pi \rfloor, \lceil e \rceil \in \mathbb{Z}$$

9

10

Simple properties

- $\lfloor -x \rfloor = -\lceil x \rceil$, hence $\lceil x \rceil = -\lfloor -x \rfloor$
- $\lfloor x + t \rfloor = \lfloor x \rfloor + t$ and $\lceil x + t \rceil = \lceil x \rceil + t$, for all $t \in \mathbb{Z}$

Fact

Let $k, m, n \in \mathbb{Z}$ such that $k > 0$ and $m \geq n$. The number of multiples of k in the interval $[n, m]$ is

$$\left\lfloor \frac{m}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor$$

11

12

Exercise

1.1.4

(b) $2 \lfloor 0.6 \rfloor - \lfloor 1.2 \rfloor = -1$

$$2 \lceil 0.6 \rceil - \lceil 1.2 \rceil = 0$$

(d) $\lceil \sqrt{3} \rceil - \lfloor \sqrt{3} \rfloor = 1$; the same for every non-integer

1.1.19(a)

Give x, y s.t. $\lfloor x \rfloor + \lfloor y \rfloor < \lfloor x + y \rfloor$

$$\lfloor 3\pi \rfloor + \lfloor e \rfloor = 9 + 2 = 11 < 12 = \lfloor 9.42\dots + 2.71\dots \rfloor = \lfloor 3\pi + e \rfloor$$

Divisibility

Exercise

1.1.4

$$(b) \ 2 \lfloor 0.6 \rfloor - \lfloor 1.2 \rfloor = -1$$

$$2 \lceil 0.6 \rceil - \lceil 1.2 \rceil = 0$$

$$(d) \ \lceil \sqrt{3} \rceil - \lfloor \sqrt{3} \rfloor = 1; \text{ the same for every non-integer}$$

1.1.19(a)

Give x, y s.t. $\lfloor x \rfloor + \lfloor y \rfloor < \lfloor x + y \rfloor$

$$\lfloor 3\pi \rfloor + \lfloor e \rfloor = 9 + 2 = 11 < 12 = \lfloor 9.42 \dots + 2.71 \dots \rfloor = \lfloor 3\pi + e \rfloor$$

Definition

Let $m, n \in \mathbb{Z}$.

' $m|n$ ' — m is a **divisor** of n , defined by $n = k \cdot m$ for some $k \in \mathbb{Z}$
Also stated as: ' n is divisible by m ', ' m divides n ', ' n multiple of m '

$m \nmid n$ — negation of $m|n$

Notion of divisibility applies to all integers (positive, negative, zero)

$1|m$, $-1|m$, $m|m$, $m|-m$, for every m

$n|0$ for every n ; $0 \nmid n$ except $n = 0$

Definition

$r = n \bmod m$ — **remainder** when n divided by m ,
defined by $n = k \cdot m + r$ for $0 \leq r < m$

$m|n$ if, and only if, $n \bmod m = 0$

13

14

Definition

$\gcd(m, n)$ — **Greatest common divisor** of m and n

$\text{lcm}(m, n)$ — **Least common multiple** of m and n

Numbers > 1 divisible only by 1 and itself are called **prime**.

Numbers m, n such that $\gcd(m, n) = 1$ are called **relatively prime**.

$\gcd(m, n)$ and $\text{lcm}(m, n)$ are always taken as positive, even if m or n is negative.

$$\gcd(-4, 6) = \gcd(4, -6) = \gcd(-4, -6) = \gcd(4, 6) = 2$$

$$\text{lcm}(-5, -5) = \dots = 5$$

Note also that $\gcd(0, 0)$ is undefined (why?)

NB

Number theory (the study of prime numbers, divisibility etc.) is important in cryptography, for example.

Absolute Value

Definition

$$|x| = \begin{cases} x & , \text{ if } x \geq 0 \\ -x & , \text{ if } x < 0 \end{cases}$$

Fact

$$\gcd(m, n) \cdot \text{lcm}(m, n) = |m| \cdot |n|$$

15

16

Exercise

1.2.2 True or False. Explain briefly.

- (a) $n|1$
- (b) $n|n$
- (c) $n|n^2$

1.2.7(b) $\gcd(0, n) \stackrel{?}{=}$

1.2.12 Can two even integers be relatively prime?

1.2.9 Let m, n be positive integers.

- (a) What can you say about m and n if $\text{lcm}(m, n) = m \cdot n$?
- (b) What if $\text{lcm}(m, n) = n$?

Exercise

1.2.2 True or False. Explain briefly.

- (a) $n|1$ — only if $n = 1$ (for $n \in \mathbb{Z}$ also $n = -1$)
- (b) $n|n$ — always
- (c) $n|n^2$ — always

1.2.7(b) $\gcd(0, n) = |n|$ (provided $n \neq 0$)

1.2.12 Can two even integers be relatively prime? No. (why?)

1.2.9 Let m, n be positive integers.

- (a) What can you say about m and n if $\text{lcm}(m, n) = m \cdot n$?
They must be relatively prime since always $\text{lcm}(m, n) = \frac{mn}{\gcd(m, n)}$
- (b) What if $\text{lcm}(m, n) = n$?
 m must be a divisor of n

Euclid's gcd Algorithm

$$f(m, n) = \begin{cases} m & \text{if } m = n \\ f(m - n, n) & \text{if } m > n \\ f(m, n - m) & \text{if } m < n \end{cases}$$

Fact

For $m > 0, n > 0$ the algorithm always terminates. (Proof?)

Fact

For $m, n \in \mathbb{Z}$, if $m > n$ then $\gcd(m, n) = \gcd(m - n, n)$

Proof.

For all $d \in \mathbb{Z}$, ($d|m$ and $d|n$) if, and only if, ($d|m - n$ and $d|n$):
 $\Rightarrow$: if $d|m$ and $d|n$ then $m = a \cdot d$ and $n = b \cdot d$, for some a, b
 then $m - n = (a - b) \cdot d$, hence $d|m - n$
 $\Leftarrow$: if $d|m - n$ and $d|n$ then ... $d|m$ (why?)

Sets

What are Sets

A set is a collection of elements. Sets are typically described by:

(a) Explicit enumeration of their elements

$$\begin{aligned} S_1 &= \{a, b, c\} = \{a, a, b, b, b, c\} \\ &= \{b, c, a\} = \dots \quad \text{three elements} \\ S_2 &= \{a, \{a\}\} \quad \text{two elements} \\ S_3 &= \{a, b, \{a, b\}\} \quad \text{three elements} \\ S_4 &= \{\} \quad \text{zero elements} \\ S_5 &= \{\{\}\} \quad \text{one element} \\ S_6 &= \{\{\}, \{\{\}\}\} \quad \text{two elements} \end{aligned}$$

Definition

Number of elements in a set X (various notations):
 $|X| = \#(X) = \text{card}(X)$

(b) Specifying the properties their elements must satisfy; the elements are taken from some 'universal' domain.

A typical description involves a **logical** property $P(x)$

$$S = \{x : x \in X \text{ and } P(x)\} = \{x \in X : P(x)\}$$

Example

$$S = \{n \in \mathbb{N} : 2|n\} \quad \text{— all even natural numbers } S = \{2, 4, 6, \dots\}$$

Definition

$x \in S$ — object x is an **element of** (or: **belongs to**) set S

Empty set $\{\}$ (alternative notation: $\emptyset$)

We distinguish between an element and the set comprising this single element. Thus always $a \neq \{a\}$.

21

22

Sets of Numbers

Natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$

Positive integers $\mathbb{P} = \{1, 2, \dots\}$

Integers $\mathbb{Z} = \{\dots, -n, -(n-1), \dots, -1, 0, 1, 2, \dots\}$

Also common notations for $\mathbb{P}$ are: $\mathbb{N}_{>0} = \mathbb{Z}_{>0} = \mathbb{N} \setminus \{0\}$

Rational numbers (fractions) $\mathbb{Q} = \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}$

Real numbers (decimal or binary expansions) $\mathbb{R}$

$$r = a_1 a_2 \dots a_k . b_1 b_2 \dots$$

In $\mathbb{P}$, $\mathbb{N}$ and $\mathbb{Z}$, different symbols denote different numbers

In $\mathbb{Q}$ and $\mathbb{R}$, the same number can be represented differently

Exercise

1.3.2 Find the cardinalities of sets

$$\textcircled{1} \quad \left| \left\{ \frac{1}{n} \in \mathbb{Q} : n \in \{1, 2, 3, 4\} \right\} \right| \stackrel{?}{=}$$

$$\textcircled{2} \quad \left| \left\{ n^2 - n \in \mathbb{N} : n \in \{0, 1, \dots, 4\} \right\} \right| \stackrel{?}{=}$$

$$\textcircled{3} \quad \left| \left\{ \frac{1}{n^2} : n \in \mathbb{P} \text{ and } 2|n \text{ and } n < 11 \right\} \right| \stackrel{?}{=}$$

$$\textcircled{4} \quad \left| \left\{ 2 + (-1)^n : n \in \mathbb{N} \right\} \right| \stackrel{?}{=}$$

23

24

Set Operations

Exercise

1.3.2 Find the cardinalities of sets

- ① $|\{ \frac{1}{n} \in \mathbb{Q} : n \in \{1, 2, 3, 4\} \}| = 4$ — four 'indices', no repetitions of values
- ② $|\{ n^2 - n \in \mathbb{N} : n \in \{0, 1, \dots, 4\} \}| = 4$ — one 'repetition' of value
- ③ $|\{ \frac{1}{n^2} : n \in \mathbb{P} \text{ and } 2|n \text{ and } n < 11 \}| = 5$
- ④ $|\{ 2 + (-1)^n : n \in \mathbb{N} \}| = 2$ — what are the two elements?

Definition

Union $A \cup B$; Intersection $A \cap B$

Note that there is a correspondence between set operations and logical operators (to be discussed in Week 2):

If A is the set of all elements x that satisfy property $a(x)$ and B is the set of all elements x that satisfy property $b(x)$ then

$A \cup B$ — elements that satisfy a or b

$A \cap B$ — elements that satisfy a and b

Definition

We say that A, B are **disjoint** if $A \cap B = \emptyset$

25

26

Other set operations

- $A \setminus B$ — **difference**, set difference, relative complement
It corresponds (logically) to: a but not b
- $A \oplus B$ — **symmetric difference**

$$A \oplus B \stackrel{\text{def}}{=} (A \setminus B) \cup (B \setminus A)$$

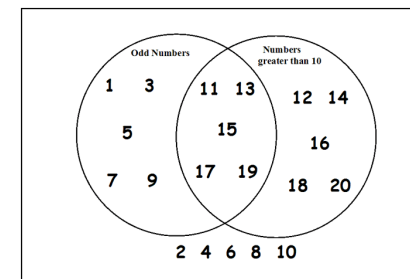
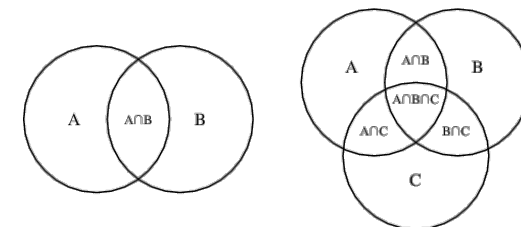
It corresponds to: a and not b or b and not a ; also known as **xor (exclusive or)**

- A^c — set **complement** w.r.t. the 'universe'
It corresponds to: not a

27

Venn Diagrams

p23–26: are a simple graphical tool to reason about the algebraic properties of set operations.



28

Laws of Set Operations

Commutativity	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distribution	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Idempotence	$A \cup A = A$ $A \cap A = A$
Identity	$A \cup \emptyset = A$ $A \cap \emptyset = \emptyset$
Double Complementation	$(A^c)^c = A$
De Morgan laws	$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

Subsets

Definition

$S \subseteq T$ — S is a **subset** of T ; includes the case of $T \subseteq T$

- all elements of S are also elements of T

$S \subset T$ — a **proper** subset: $S \subseteq T$ and $S \neq T$

NB

An element of a set and a subset of that set are different concepts

$$a \in \{a, b\}, \quad a \not\subseteq \{a, b\}; \quad \{a\} \subseteq \{a, b\}, \quad \{a\} \notin \{a, b\}$$

Fact

$$A \cup B = B \Leftrightarrow A \subseteq B \quad A \cap B = B \Leftrightarrow A \supseteq B$$

$$\emptyset \subseteq X \text{ for all sets } X$$

Power Set

Definition

Power set $\text{Pow}(X) = \{ A : A \subseteq X \}$

Fact

$$\text{Always } |\text{Pow}(X)| = 2^{|X|}$$

Examples

$$|\emptyset| = 0 \quad \text{Pow}(\emptyset) = \{\emptyset\} \quad |\text{Pow}(\emptyset)| = 1$$

$$\text{Pow}(\text{Pow}(\emptyset)) = \{\emptyset, \{\emptyset\}\} \quad |\text{Pow}(\text{Pow}(\emptyset))| = 2 \quad \dots$$

$$|\{a\}| = 1 \quad \text{Pow}(\{a\}) = \{\emptyset, \{a\}\} \quad |\text{Pow}(\{a\})| = 2 \quad \dots$$

Note the **type**: $\text{Pow}(X)$ is always a **set** of **sets**

Exercise

$$1.4.4 \quad \Sigma = \{a, b\}$$

(d) All subsets of Σ ?

$$(e) |\text{Pow}(\Sigma)| \stackrel{?}{=}$$

$$1.4.7 \quad A \oplus A \stackrel{?}{=}, \quad A \oplus \emptyset \stackrel{?}{=}$$

$$1.4.8 \quad \text{Relate the cardinalities } |A \cup B|, |A \cap B|, |A \setminus B|, |A \oplus B|, |A|, |B|$$

Exercise

1.4.4 $\Sigma = \{a, b\}$

(d) All subsets of Σ ? $\emptyset, \{a\}, \{b\}, \{a, b\}$

(e) $|\text{Pow}(\Sigma)| = 4$

1.4.7 $A \oplus A = \emptyset, \quad A \oplus \emptyset = A$ for all A

1.4.8 Relate the cardinalities:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

hence $|A \cup B| + |A \cap B| = |A| + |B|$

$$|A \setminus B| = |A| - |A \cap B|$$

$$|A \oplus B| = |A| + |B| - 2|A \cap B|$$

- Number sets and their containments

$$\mathbb{P} \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

- Derived sets of positive numbers

$$\mathbb{N}_{>0} = \mathbb{Z}_{>0} = \{n \in \mathbb{Z} : n \geq 1\} \subset \mathbb{Q}_{>0} = \{\frac{m}{n} : m, n \in \mathbb{Z}_{>0}\} \subset \mathbb{R}_{>0}$$

- Derived sets of integers

$$2\mathbb{Z} = \{2x : x \in \mathbb{Z}\}$$

the even numbers

$$3\mathbb{Z} + 1 = \{3x + 1 : x \in \mathbb{Z}\}$$

33



34



Intervals

Definition

Intervals of numbers (applies to any type)

$$[a, b] = \{x : a \leq x \leq b\}; \quad (a, b) = \{x : a < x < b\}$$

$$[a, b] \supseteq [a, b], \quad (a, b) \supseteq (a, b)$$

NB

$(a, a) = (a, a] = [a, a) = \emptyset$; however $[a, a] = \{a\}$.

Intervals of $\mathbb{P}, \mathbb{N}, \mathbb{Z}$ are finite: if $m \leq n$

$$[m, n] = \{m, m+1, \dots, n\} \quad |[m, n]| = n - m + 1$$

Exercise

1.3.10 Number of elements in the sets

① $\{-1, 1\}$

② $[-1, 1]$

③ $(-1, 1)$

35



36



Exercise

1.3.10 Number of elements in the sets

- ① $\{-1, 1\}$ — 2
- ② $[-1, 1]$ — 3 (if over $\mathbb{Z}$); ∞ (if over $\mathbb{Q}$ or $\mathbb{R}$)
- ③ $(-1, 1)$ — 1 (if over $\mathbb{Z}$); ∞ (if over $\mathbb{Q}$ or $\mathbb{R}$)

Sidetrack

Proper ways to [introduce reals](#) include Dedekind cuts and Cauchy sequences, neither of which will be discussed here.

Natural numbers etc. are either axiomatised or [constructed from sets](#) ($0 \stackrel{\text{def}}{=} \{\}$, $1 \stackrel{\text{def}}{=} \{0, \{0\}\} = \{\{\}, \{\{\}\}\}, \dots$)

NB

If we need to emphasise that an object (expression, formula) is defined through an equality we use the symbol $\stackrel{\text{def}}{=}$. It denotes that the object on the left is defined by the formula/expression given on the right.

37



38



Cartesian Product

Definition

$$S \times T \stackrel{\text{def}}{=} \{ (s, t) : s \in S, t \in T \} \quad (s, t) \text{ is an } \mathbf{ordered} \text{ pair}$$

$$\times_{i=1}^n S_i \stackrel{\text{def}}{=} \{ (s_1, \dots, s_n) : s_k \in S_k, \text{ for } 1 \leq k \leq n \}$$

$$S^2 = S \times S, \quad S^3 = S \times S \times S, \dots$$

Fact

$$\emptyset \times S = \emptyset, \text{ for every } S$$

$$|S \times T| = |S| \cdot |T|, \quad |\times_{i=1}^n S_i| = \prod_{i=1}^n |S_i|$$

39



Functions

We deal with functions as a set-theoretic concept, it being a special kind of correspondence (between two sets)

$f : S \longrightarrow T$ describes pairing of the sets: it means that f assigns to every element $s \in S$ a unique element $t \in T$

To emphasise that a specific element is sent, we can write $f : x \mapsto y$, which means the same as $f(x) = y$



40

Formal Languages

Σ — **alphabet**, a finite, nonempty set

Examples (of various alphabets and their intended uses)

$\Sigma = \{a, b, \dots, z\}$ for single words (in lower case)

$\Sigma = \{\sqcup, -, a, b, \dots, z\}$ for composite terms

$\Sigma = \{0, 1\}$ for binary integers

$\Sigma = \{0, 1, \dots, 9\}$ for decimal integers

The above cases all have a natural ordering; this is not required in general, thus the set of all Chinese characters forms a (formal) alphabet.

Words

Definition

word — any finite string of symbols from Σ

empty word — λ

Examples

$\omega = aba, \omega = 01101 \dots 1$, etc.

$\text{length}(\omega)$ — # of symbols in ω

$\text{length}(aaa) = 3, \text{length}(\lambda) = 0$

The only operation on words (discussed here) is **concatenation**, written as juxtaposition $\nu\omega, \omega\nu\omega, ab\omega, \omega b\nu, \dots$

NB

$\lambda\omega = \omega = \omega\lambda$

$\text{length}(\nu\omega) = \text{length}(\nu) + \text{length}(\omega)$

Notation: Σ^k — set of all words of length k

We often identify $\Sigma^0 = \{\lambda\}, \Sigma^1 = \Sigma$

Σ^* — set of all words (of all lengths)

Σ^+ — set of all nonempty words (of any positive length)

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots; \quad \Sigma^{\leq n} = \bigcup_{i=0}^n \Sigma^i$$

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \dots = \Sigma^* \setminus \{\lambda\}$$

A **language** is a subset of Σ^* . Typically, only the subsets that can be formed (or described) according to certain rules are of interest. Such a collection of 'descriptive/formative' rules is called a **grammar**.

Examples: Programming languages, Database query languages

Formal Logic

Examples

1.3.10 Number of elements in the sets (cont'd)

(e) Σ^* where $\Sigma = \{a, b, c\}$ — $|\Sigma^*| = \infty$

(f) $\{\omega \in \Sigma^* : \text{length}(\omega) \leq 4\}$ where $\Sigma = \{a, b, c\}$

$$|\Sigma^{\leq 4}| = 3^0 + 3^1 + \dots + 3^4 = \frac{3^5 - 1}{3 - 1} = \frac{243 - 1}{2} = 121$$

Elementary Logic

Exercise

Claim:

A *necessary* condition for the program to terminate is to input a positive number.

Suppose you want to formally verify this claim. Which would be the correct logical way of formalising and proving this?

- $\text{Terminates} \Rightarrow \text{Positive_Input}$ **correct**
- $\text{Positive_Input} \Rightarrow \text{Terminates}$

Elementary Logic

Exercise

Claim:

A *necessary* condition for the program to terminate is to input a positive number.

Suppose you want to formally verify this claim. Which would be the correct logical way of formalising and proving this?

- $\text{Terminates} \Rightarrow \text{Positive_Input}$ **correct**
- $\text{Positive_Input} \Rightarrow \text{Terminates}$

Proofs

A **mathematical proof** of a proposition p is a chain of logical deductions leading to p from a base set of axioms.

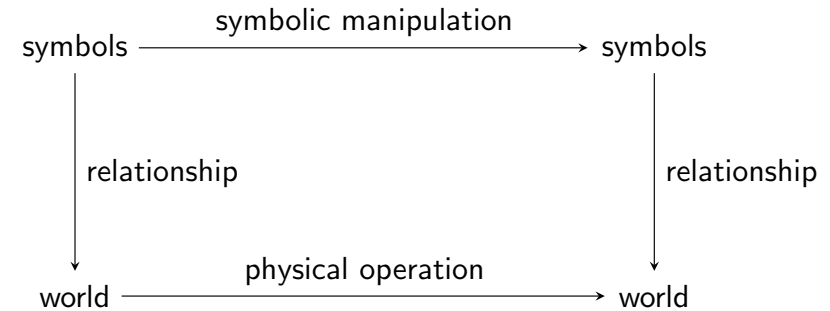
Example

Proposition: Every group of 6 people includes a group of 3 who each have met each other or a group of 3 who have not met a single other person in that group.

Proof: by case analysis.

But what are propositions, logical deductions, and axioms? And what is a sound case analysis?

The Real World vs Symbols



NB

"Essentially, all models are wrong. But some are useful."

(George Box)

The main relationship between symbols and the world of concern in logic is that of a *sentence of a language* being *true* in the world. A sentence of a natural language (like English, Cantonese, Warlpiri) is *declarative*, or a **proposition**, if it can be meaningfully be said to be either true or false.

Examples

- There are 35 Aboriginal language groups in New South Wales.
- A square root of 16 is 4.
- Euclid's program gets stuck in an infinite loop if you input 0.
- Whatever list of numbers you give as input to this program, it outputs the same list but in increasing order.
- $x^n + y^n = z^n$ has no nontrivial integer solutions for $n > 2$.

The following are *not* declarative sentences of English:

- Gubble gimble goo
- For Pete's sake, take out the garbage!
- Did you watch MediaWatch last week?
- Please waive the prerequisites for this subject for me.

Declarative sentences in natural languages can be *compound* sentences, built out of other sentences.

Propositional Logic is a formal representation of some constructions for which the truth value of the compound sentence can be determined from the truth value of its components.

Examples

- Lists L and M contain the same elements *and* M is ordered.
- Either you have a passport *or* you cannot travel abroad.
- *It is not the case that* this program always halts.

Not all constructions of natural language are truth-functional:

- *Trump believes that* Iran is developing nukes.
- This program always halts *because* it contains no loops.
- The disk crashed *after* I saved my file.

NB

Various **modal logics** extend classical propositional logic to represent, and reason about, these and other constructions.

53

Navigation icons

54

Navigation icons

Propositional Logic

Definition

symbol	text
$\wedge$	"and", "but", ":", ":",
$\vee$	"or", "either ... or ..."
$\neg$	"not", "it is not the case that"

Truth tables:

A	B	$A \wedge B$
F	F	F
F	T	F
T	F	F
T	T	T

A	B	$A \vee B$
F	F	F
F	T	T
T	F	T
T	T	T

A	$\neg A$
F	T
T	F

55

Navigation icons

Applications I: Program Logic

Example

if $x > 0$ or $(x \leq 0$ and $y > 100)$:

Let $p \stackrel{\text{def}}{=} (x > 0)$ and $q \stackrel{\text{def}}{=} (y > 100)$

$p \vee (\neg p \wedge q)$

p	q	$p \vee (\neg p \wedge q)$
F	F	F
F	T	T
T	F	T
T	T	T

This is equivalent to $p \vee q$. Hence the code can be simplified to

if $x > 0$ or $y > 100$:

Navigation icons

56

Somewhat more controversially, consider the following constructions:

- if A then B
- A only if B
- B if A
- A implies B
- it follows from A that B
- whenever A, B
- A is a sufficient condition for B
- B is a necessary condition for A

Each has the property that if true, and A is true, then B is true.

Example

If you are from England *then* you are from the UK.

Definition

We can *approximate* the English meaning of these by “not (A and not B)”, written $A \Rightarrow B$, which has the following truth table:

A	B	$A \Rightarrow B$
F	F	T
F	T	T
T	F	F
T	T	T

While only an approximation to the English, 100+ years of experience have shown this to be adequate for capturing *mathematical reasoning*.

(Moral: mathematical reasoning does not need all the features of English.)

Converse of $A \Rightarrow B$ is $B \Rightarrow A$

Exercise

LLM: Problem 3.2

- p = “you get an HD on your final exam”
- q = “you do every exercise in the book”
- r = “you get an HD in the course”

Translate into logical notation:

- (a) You get an HD in the course although you do not do every exercise in the book.
- (c) To get an HD in the course, you must get an HD on the exam.
- (d) You get an HD on your exam, but you don’t do every exercise in this book; nevertheless, you get an HD in this course.

Exercise

LLM: Problem 3.2

- p = “you get an HD on your final exam”
- q = “you do every exercise in the book”
- r = “you get an HD in the course”

Translate into logical notation:

- (a) You get an HD in the course although you do not do every exercise in the book. $r \wedge \neg q$
- (c) To get an HD in the course, you must get an HD on the exam. $r \Rightarrow p$
- (d) You get an HD on your exam, but you don’t do every exercise in this book; nevertheless, you get an HD in this course. $p \wedge \neg q \wedge r$

Unless

A **unless** B can be approximated as $\neg B \Rightarrow A$

Example

"I go swimming unless it rains" = "If it is not raining I go swimming"

Correctness of the translation is perhaps easier to see in:

"I don't go swimming unless the sun shines" = "If the sun does not shine then I don't go swimming."

NB

"I go swimming unless it rains, but sometimes I swim even though it is raining" makes sense, so the translation of "A unless B" should not imply $B \Rightarrow \neg A$.

Just in case

A **just in case** B usually means A if, and only if, B; written $A \Leftrightarrow B$

Examples

"The program terminates just in case you input a positive number"
= "The program terminates if, and only if, the input is positive"

"I will go swimming just in case I won't play soccer"
= "If I play soccer I will not go swimming and vice versa"

Definition

A	B	$A \Leftrightarrow B$
F	F	T
F	T	F
T	F	F
T	T	T

Same as $(A \Rightarrow B) \wedge (B \Rightarrow A)$

61

62

The Formal Language of Propositional Logic

Let $Prop = \{p, q, r, \dots\}$ be a set of basic propositional letters.
Consider the *alphabet* $\Sigma = Prop \cup \{\top, \perp, \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow, (,)\}$

Definition

The set of **formulae of propositional logic** is the smallest set of words over Σ such that

- $\top$, $\perp$ and all elements of $Prop$ are formulae
- If ϕ is a formula, then so is $\neg\phi$
- If ϕ and ψ are formulae, then so are $(\phi \wedge \psi)$, $(\phi \vee \psi)$, $(\phi \Rightarrow \psi)$, and $(\phi \Leftrightarrow \psi)$.

$\top$: formula that is always true; $\perp$: formula that is always false

Convention: we often drop parentheses when there is no ambiguity.

$\neg$ binds more tightly than $\wedge$ and $\vee$, which in turn bind more tightly than $\Rightarrow$ and $\Leftrightarrow$.

Finally... Some Supplementary Exercises

Exercise

1.8.2(b) (supp) When is $(A \setminus B) \setminus C = A \setminus (B \setminus C)$?

1.8.9 (supp) How many third powers are $\leq 1,000,000$ and end in 9? (Solve without calculator!)

63

64

Exercise

1.8.2(b) (supp) When is $(A \setminus B) \setminus C = A \setminus (B \setminus C)$?

From Venn diagram

$$(A \setminus B) \setminus C = A \cap B^c \cap C^c; \quad A \setminus (B \setminus C) = (A \cap B^c) \cup (A \cap C).$$

Equality would require that $A \cap C \subseteq A \cap B^c \cap C^c$; however, these two sets are disjoint, thus $A \cap C = \emptyset$ is a necessary condition for the equality.

One verifies that $A \cap C = \emptyset$ is also a sufficient condition and that, in this case, both set expressions simplify to $A \setminus B$.

1.8.9 (supp) How many third powers are $\leq 1,000,000$ and end in 9? (Solve without calculator!)
 $n^3 = 9 \pmod{10}$ only when $n = 9 \pmod{10}$, and $n^3 \leq 1,000,000$ when $n \leq 100$. Hence all such n are $9, 19, \dots, 99$.
 Try the same question for n^4 .

- Notation for numbers
 $\lfloor m \rfloor, \lceil m \rceil, m|n, n \bmod m, |a|, [a, b], (a, b), \gcd, \text{lcm}$
- Sets and set operations
 $|A|, \in, \cup, \cap, \setminus, \oplus, A^c, \text{Pow}(A), \subseteq, \subset, \times$
- Formal languages: alphabets and words
 $\lambda, \Sigma^*, \Sigma^+, \Sigma^1, \Sigma^2, \dots, \Sigma^{\leq k}$
- Language of propositional logic
 $\wedge, \vee, \neg, \Rightarrow, \Leftrightarrow, \top, \perp, \text{truth tables}$

Coming up ...

- Ch. 2, Sec. 2.2-2.5 (Propositional calculus, Proofs)
- Ch. 10, Sec. 10.1-10.5 (Boolean algebra)